

Distributed Secure Surrounding Control for Multiple USVs Against Deception Attacks: A Stackelberg Game Approach With Reinforcement Learning

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Abstract—This article investigates the Stackelberg game-based distributed secure surrounding control (SSC) problem for multiple unmanned surface vehicles (USVs) with unknown dynamics under deception attacks (DAs). The proposed scheme is rooted in the Stackelberg game with reinforcement learning (RL), in which the attacker and controller respectively play the roles of the follower and leader, making sequential decisions of the DA and the SSC. Specifically, a distributed target estimator is established to access the target position. By utilizing this estimated position to formulate an intermediate control law, the target surrounding scenario is effectively transformed into the Stackelberg game-solving problem. The RL approach with a neural networks-based actor-critic learning structure is deployed to directly derive the distributed optimal SSC and DA policies from the Bellman error, whilst learning the unknown dynamics of USVs. Moreover, a value function decomposition technique is applied optimally for using the designed control parameters, thereby accelerating the acquisition of optimal policies. A rigorous theoretical analysis is employed to ensure the closed-loop stability of multiple USVs. Simulations are provided to validate the effectiveness of the proposed distributed SSC scheme for multiple USVs.

Index Terms—Distributed secure surrounding control, multiple unmanned surface vehicles, deception attacks, Stackelberg game, reinforcement learning.

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I. INTRODUCTION

MULTIPLE USVs with exceptional flexibility, robust adaptability, and economical maintenance have found successful applications in many important civil and military domains, including formation escort, marine environment detection, and collaborative disaster search [1], [2], [3]. A brief overview of collaborative control for USVs is provided in [4]. Numerous investigations have been conducted in the realm of collaborative control for USVs, with a concentration on cooperative target tracking [5], leader-follower formation [6], and path planning [7]. Particularly, the distributed target surrounding control of USVs has recently garnered significant interest [8], [9], [10]. By utilizing extended state observers and data-driven fuzzy predictors, the distributed surrounding control of multiple USVs with unknown dynamics is implemented, all without relying on global information [9]. Thus, investigating the distributed target surrounding control of multiple USVs with unknown dynamics is challenging yet crucial.

Distributed target surrounding control is a technique that utilizes local information to guide USVs in surrounding targets with the specified formation while maintaining desired distances [11]. This approach has been successfully used in applications like maritime patrols, rescue, and maritime cluster operations [12], [13]. The distributed estimator-based equal-surrounding control is developed to achieve the uniform target surrounding configuration, validated on USVs [12]. However, multiple USVs encounter challenges in network security, including malicious cyber-attacks [14] and stealthy attacks on remote actuators and sensors [15]. The injection of false data into the communication network from such attacks can lead to task execution failures and even system crashes. Consequently, extensive research has been conducted on distributed control of USVs under malicious attacks [16], [17], [18]. The secure collaborative path tracking method with the extended state observer is developed to enable multiple USVs to withstand malicious cyber-physical attacks [17]. Specifically, the distributed control of multiple USVs under random attacks [19], denial-of-service attacks [20], and DAs [21] has been investigated. It should be pointed out that practical scenarios involve not only data injection attacks with single constants and bounded random functions but also targeted attacks based on obtained state information. However, conventional methods for resisting attacks fall short of directly

addressing the distributed surrounding problem of multiple USVs under covert and targeted attacks orchestrated by DAs. The interactive decision-making between the defender and attacker is explored in the Stackelberg game framework to counter malicious attacks aiming to degrade system performance [22]. Hence, a necessary yet challenging task is to devise a novel distributed SSC scheme for multiple USVs under DAs.

The Stackelberg game, a dynamic paradigm involving two entities termed the leader and the follower, operates under the sequential decision-making framework [22], [23], [24]. Herein, the leader initiates decision-making, prompting reactions from the followers. Meanwhile, the leader strategically accounts for the optimal decision considering the best response from its followers. This characteristic feature underpins its utility across diverse applications, including but not limited to smart grids [25] and the design of defense strategies [23]. The Stackelberg differential graph game is explored to address distributed optimal fault estimation and fault-tolerant control challenges, considering potential bidirectional interactions between the fault and observer [26]. In practical scenarios, the unpredictable nature of marine environments often results in USVs encountering challenges such as weak communication signals, low positioning accuracy, and limited perception range [3], [27], [28], [29]. These factors hinder the acquisition of accurate dynamics for USVs, thus presenting significant obstacles in solving the Stackelberg game-based distributed SSC problem. To tackle this challenge, RL, serving as a goal-oriented learning tool, offers an effective solution for addressing optimal control problems [30], [31], [32]. This approach eliminates the need for solving the Hamilton-Jacobi (HJ) equation without an analytical form. Specifically, a data-driven performance-prescribed RL control method is developed to directly optimize the optimal control policy derived from the Bellman error [32]. Additionally, actor-critic neural networks (NNs) are constructed to solve the HJ equation corresponding to the USVs with unknown dynamics, aiming to achieve adaptive optimal tracking control [33]. Hence, it is critically desired to develop a Stackelberg game scheme with RL to address the distributed SSC problem of multiple USVs with unknown dynamics under DAs.

Given the research gaps identified above, this article employs the RL approach to design the distributed SSC scheme for multiple USVs with unknown dynamics within the Stackelberg game framework, addressing targeted DAs that aim at maliciously disrupting the control objective. Furthermore, the principal contributions can be outlined as follows:

- 1) A Stackelberg game-based distributed SSC scheme is proposed to derive the distributed optimal SSC and DA policies for making effective sequential decisions, aiming to deal with more covert and targeted DAs. Especially, with the proposed scheme, distributed target surrounding control for multiple USVs against such attacks can be achieved without needing the global target information, target velocity, and vehicle dynamics.
- 2) Under the premise of accessibility to the Nash equilibrium point, the RL-based Stackelberg game solution with an additional NN is constructed. This method directly optimizes the Bellman error through online learning to obtain the available optimal policies for the attacker and controller

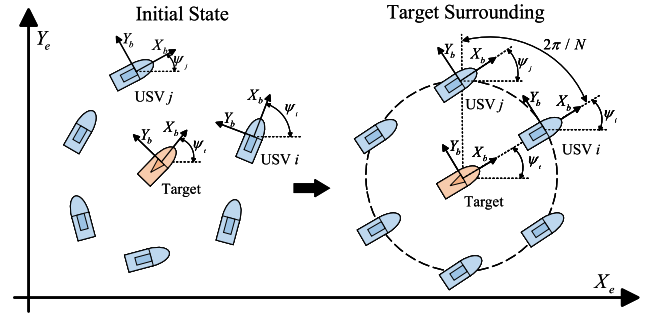


Fig. 1. Cooperative surrounding scenario in both earth- and body-fixed frame.

that satisfy the coupled HJ equation, eliminating the need for prior knowledge of USV dynamics.

- 3) A value function decomposition technique is established, which improves the learning ability of distributed SSC policy for multiple USVs by fully embedding the control parameters designed by the controller, thereby providing potent compensation for unknown dynamics and external disturbances. Furthermore, stability analysis is conducted to guarantee the boundedness of all signals, even in the presence of DAs, unknown target velocity, and unknown vehicle dynamics.

The remainder of this article is structured as follows: Section II presents preliminaries and problem formulation. The design of distributed optimal SSC and DA policies with Stackelberg game is introduced in Section III. Section IV provides a discussion of RL-based Stackelberg game solution, along with stability analysis. In Section V and Section VI, simulation results and conclusions are provided, respectively.

Notations: $\mathbb{R}$ denotes the sets of real numbers, $\text{diag}\{\cdot\} > 0$ denotes the diagonal positive definite matrix, I_n represents the $n \times n$ identity matrix, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ represent the minimum and maximum eigenvalues, and $\max\{\cdot\}$ represents the maximum elements. $\|x\|$ is the Euclidean norm for the vector x , $\text{tr}(X)$ and $\|X\|_F$ are the trace and Frobenius norm for the matrix X , respectively.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Graph Theory

Consider a grouping of N USVs, wherein the configuration of their interaction topology is denoted by a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V} = \{1, \dots, N\}$ being the node set comprising USVs and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denoting the edge set portraying communication links interconnecting the USVs. Let a set $\mathcal{N}_i = \{j | (j, i) \in \mathcal{E}\}$ denotes the neighbors of the i -th USV ($i \in \mathcal{V}$), the adjacency weight matrix can be denoted by $\mathcal{A}_i = [a_{ij}]_{n \times n}$. If $(i, j) \in \mathcal{E}$, then $a_{ij} = 1$; otherwise, $a_{ij} = 0$. The Laplace matrix associated with the graph $\mathcal{G}$, represented as $\mathcal{L}$, is defined as $\mathcal{L} = \mathcal{D} - \mathcal{A}$, where $\mathcal{D} = \text{diag}[d_1, \dots, d_N]$ with $d_i = \sum_{j=1}^N a_{ij}$ for $i = 1, \dots, N$.

B. Problem Formulation

As illustrated in Fig. 1, the scenario scrutinized in this article involves N USVs and a target agent. The USVs are directed to encircle the moving target agent. The kinematics of the target

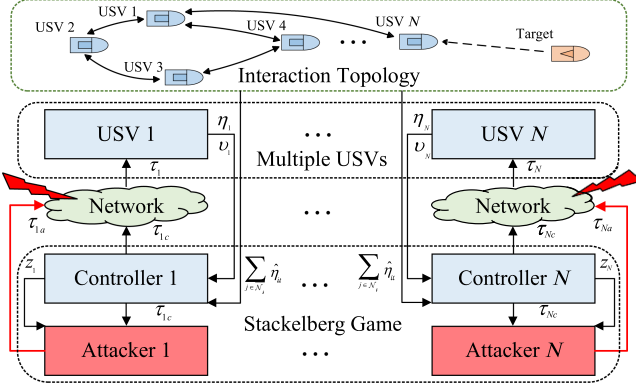


Fig. 2. Structure with Stackelberg game, USVs, and interaction topology.

agent can be characterized as

$$\dot{\eta}_t = \mathcal{R}(\eta_t) v_t \quad (1)$$

where $\eta_t = [x_t, y_t, \psi_t]^T \in \mathbb{R}^3$ is the position/heaving vector, with x_t and y_t being the target's positions and ψ_t denoting yaw angle in the earth-fixed frame, $v_t = [u_t, v_t, r_t]^T \in \mathbb{R}^3$ denotes the unknown velocity vector including the surge velocity u_t , sway velocity v_t , and yaw rate r_t in the body-fixed frame, and the rotational matrix $\mathcal{R}(\eta_t) \in \mathbb{R}^{3 \times 3}$ is denoted as $\mathcal{R}(\eta_t) = [\cos \psi_t, -\sin \psi_t, 0; \sin \psi_t, \cos \psi_t, 0; 0, 0, 1]$.

Assumption 1: The target's acceleration is bounded, i.e., $\|\dot{v}_t\| \leq \bar{v}_t^*$ with $\bar{v}_t^*$ denoting an unknown positive constant.

The kinematics and dynamics of the i -th USV, denoted as USV i , can be expressed as

$$\begin{cases} \dot{\eta}_i = \mathcal{R}(\eta_i) v_i \\ M_i \dot{v}_i + f_i(\eta_i, v_i) = \tau_i + \tau_{id} \end{cases} \quad (2)$$

where $\eta_i = [x_i, y_i, \psi_i]^T \in \mathbb{R}^3$ represents the USV's position/heaving vector with positions x_i , y_i and yaw angle ψ_i , $v_i = [u_i, v_i, r_i]^T \in \mathbb{R}^3$ is the velocity vector with the surge velocity u_i , sway velocity v_i , and yaw angular velocity r_i , and $\mathcal{R}(\eta_i) = [\cos \psi_i, -\sin \psi_i, 0; \sin \psi_i, \cos \psi_i, 0; 0, 0, 1]$ is the rotational matrix. $M_i \in \mathbb{R}^{3 \times 3}$ denotes the inertial mass matrix, and $f_i(\eta_i, v_i) = C_i(v_i)v_i + D_i(v_i)v_i + \Delta_i(\eta_i, v_i)$ represents unknown nonlinear dynamics with $C_i(v_i) \in \mathbb{R}^{3 \times 3}$, $D_i(v_i) \in \mathbb{R}^{3 \times 3}$ and $\Delta_i(\eta_i, v_i) \in \mathbb{R}^3$ being centripetal matrix, damping matrix, and unmodeled dynamics, respectively, and $\tau_i \in \mathbb{R}^3$ and $\tau_{id} \in \mathbb{R}^3$ denote the input signal of the actuator and bounded external disturbances, respectively.

Assuming that the data transmission from the controller module to the actuator of USV i utilizes a shared and open communication network, there exists a risk of potential attackers hijacking the network layer. As depicted in Fig. 2, the attacker aims to inject false data into the designated control signal, thereby disrupting the collaboration among multiple USVs. Consequently, the input signal of USV i under DAs is introduced as

$$\tau_i = \tau_{ic} + \tau_{ia} \quad (3)$$

where $\tau_{ic} \in \mathbb{R}^3$ is the control input to be designed and $\tau_{ia} \in \mathbb{R}^3$ represents the DA signal caused by attackers.

Assumption 2: The attackers possess the capability to obtain the system information and control objective and continuously attack all actuators with false data vectors calculated from the acquired information.

In the gaming scenario involving interaction between the attacker and the controller, the focus lies in formulating the optimal SSC policy for the USV i while concurrently devising the optimal DA policy for the attacker. The control objective in this article is to implement the distributed SSC policy under the DA policy, enabling USVs (2) with unknown dynamics to cooperatively capture and track the moving target (1) with the unknown velocity. To this end, the control objective for the USV i is mathematically described as

$$\lim_{t \rightarrow \infty} \|\eta_i - \eta_t - p_{it}\| \leq \sigma_{it} \quad (4)$$

where $p_{it} = [\rho \cos(2\pi i/N + \psi_t), \rho \sin(2\pi i/N + \psi_t), 0]^T$ with ρ representing the target surrounding radius.

Definition 1: (Best Response) The best response is a strategy that a player chooses to minimize its cost function, given the strategies of the other participants.

Definition 2: (Stackelberg Equilibrium) A unique Stackelberg equilibrium is achieved through the control pair $\{\tau_{ic}^*, \tau_{ia}^*(\tau_{ic}^*)\}$ that adheres to the following properties:

- 1) For each τ_{ia} , there always exists a unique control $\tau_{ic}^*(\tau_{ia}^*)$ on the best responses of the follower that minimizes $J_{ia}(\tau_{ic}, \tau_{ia})$, i.e., $J_{ia}(\tau_{ic}, \tau_{ia}^*(\tau_{ic})) \leq J_{ia}(\tau_{ic}, \tau_{ia}(\tau_{ic})), \forall \tau_{ic}$.
- 2) The leader aims to find the optimal solution τ_{ic}^* to minimize $J_{ic}(\tau_{ic}, \tau_{ia}^*(\tau_{ic}))$, for any pair $\{\tau_{ic}, \tau_{ia}(\tau_{ic})\}$, i.e., $J_{ic}(\tau_{ic}^*, \tau_{ia}^*(\tau_{ic}^*)) \leq J_{ic}(\tau_{ic}, \tau_{ia}^*(\tau_{ic}))$.

Remark 1: Previous studies commonly treat DAs as either constant or bounded functions [16], [17], [18], [21]. In contrast, in this article, τ_{ia} is subjectively designed to allow the controller to perform strategically dynamic adjustments based on acquired information. Thus, this attack is characterized as covert and targeted. Assumption 2 delineates the attacker's ability to access system information and compute attack signals. Since the focus of this article is on developing the distributed SSC control scheme to tackle these attack strategies, the constraints on the acquisition and utilization of system information have been appropriately relaxed.

III. DISTRIBUTED OPTIMAL SSC AND DA POLICIES WITH STACKELBERG GAME

In this section, the Stackelberg game framework is employed to formulate optimal control policies for both the attacker and the controller. Firstly, the distributed target estimator is devised, enabling each USV to perceive target position information through the communication network. Then, the attacker and controller assume the roles of the follower and the leader, providing relevant configurations based on the estimated information within the Stackelberg game framework. Finally, the distributed optimal SSC and DA policies for USVs are derived from the Stackelberg game.

A. Design of Distributed Target Estimator

In this article, the acquisition of the target pose information is restricted to some USVs, contrasting with the assumption made in [34] that requires global accessibility to the target information. To address this issue, the target velocity in the earth-fixed frame is defined as $\zeta_t = [\dot{x}_t, \dot{y}_t, \dot{\psi}_t]^\top$, and the distributed target estimator for the USV i is formulated as

$$\begin{cases} \dot{\hat{\eta}}_{it} = -c\vartheta_{t1}(\sum_{j=1}^N l_{ij}(\hat{\eta}_{it} - \hat{\eta}_{jt}) + \gamma_{it}(\hat{\eta}_{it} - \eta_t)) + \hat{\zeta}_{it} \\ \dot{\hat{\zeta}}_{it} = -c\vartheta_{t2}(\sum_{j=1}^N l_{ij}(\hat{\eta}_{it} - \hat{\eta}_{jt}) + \gamma_{it}(\hat{\eta}_{it} - \eta_t)) \end{cases} \quad (5)$$

where $\hat{\eta}_{it} = [\hat{x}_{it}, \hat{y}_{it}, \hat{\psi}_{it}]^\top$ and $\hat{\zeta}_{it} = [\hat{\dot{x}}_{it}, \hat{\dot{y}}_{it}, \hat{\dot{\psi}}_{it}]^\top$ denote the estimated target pose and velocity, respectively, c is a positive constant, and ϑ_{tj} , $j = 1, 2 \in \mathbb{R}^{3 \times 3}$ are positive definite matrices. $\gamma_{it} = 1$ (at least one γ_{it} is nonzero) indicates that the target information is obtained by USV i , while $\gamma_{it} = 0$ indicates the opposite scenario.

Define the estimated target errors as $\tilde{\eta}_{it} = \hat{\eta}_{it} - \eta_t$ and $\tilde{\zeta}_{it} = \hat{\zeta}_{it} - \zeta_t$, the time derivative of $\tilde{\eta}_{it}$, $\tilde{\zeta}_{it}$ are given by

$$\begin{cases} \dot{\tilde{\eta}}_{it} = -c\vartheta_{t1}(\sum_{j=1}^N l_{ij}(\tilde{\eta}_{it} - \tilde{\eta}_{jt}) + \gamma_{it}\tilde{\eta}_{it}) + \tilde{\zeta}_{it} \\ \dot{\tilde{\zeta}}_{it} = -c\vartheta_{t2}(\sum_{j=1}^N l_{ij}(\tilde{\eta}_{it} - \tilde{\eta}_{jt}) + \gamma_{it}\tilde{\eta}_{it}) - \dot{\tilde{\eta}}_t \end{cases} \quad (6)$$

Let $Y_t = [Y_{1t}^\top, \dots, Y_{Nt}^\top]^\top$ with $Y_{it} = [\tilde{\eta}_{it}^\top, \tilde{\zeta}_{it}^\top]^\top$, it is observed that:

$$\dot{Y}_t = (I_N \otimes A - cH_t \otimes \Theta_t C)Y_t - 1_N \otimes (B\dot{\eta}_t) \quad (7)$$

where $A = [0_3, I_3; 0_3, 0_3]^\top$, $B = [0_3, I_3]^\top$, $C = [I_3, 0_3]$, $\Theta_t = [\vartheta_{t1}, \vartheta_{t2}]^\top$, $H_t = \mathcal{L} + \mathcal{Y}_t$ with $\mathcal{Y}_t = \text{diag}\{\gamma_{1t}, \dots, \gamma_{Nt}\}$, and $1_N = [1, \dots, 1]^\top$. Assuming the existence of a positive definite symmetric matrix P_t such that $P_t A + A^\top P_t - 2C^\top C + Q \leq 0$ with Q being a positive definite matrix.

To evaluate the interaction between the attacker and the controller of the USV i in the game, a target surrounding error $e_i = \eta_i - \hat{\eta}_{it} - \hat{p}_{it}$ is introduced using the estimated target pose with $\hat{p}_{it} = [\rho \cos(2\pi i/N + \hat{\psi}_{it}), \rho \sin(2\pi i/N + \hat{\psi}_{it}), 0]^\top$. And then, the time derivative of e_i is

$$\dot{e}_i = \mathcal{R}(\eta_i)v_i - \Lambda_i \hat{\zeta}_{it} \quad (8)$$

where $\Lambda_i = [1, 0, \rho \sin(\frac{2\pi i}{N} + \hat{\psi}_t); 0, 1, \rho \cos(\frac{2\pi i}{N} + \hat{\psi}_t); 0, 0, 1]$.

B. Stackelberg Game Setting

To ensure the convergence of the target surrounding error e_i and streamline the control design process, the intermediate surrounding control law is formulated using (8) as

$$v_i^d = \mathcal{R}(\eta_i)^{-1}(-K_{ei}e_i + \Lambda_i \hat{\zeta}_{it}) \quad (9)$$

where $K_{ei} \in \mathbb{R}^{3 \times 3}$ is a positive definite matrix.

The discrepancy between the velocity v_i and the control law v_i^d is pivotal for the convergence of the target surrounding error e_i . To this end, the velocity tracking error is denoted as $z_i = v_i - v_i^d$. Then, by taking the time derivative of z_i and incorporating it with the dynamics (2) of the USV i , the velocity error dynamics under DAs can be obtained as

$$\dot{z}_i = h_i(z_i) + g_i(\tau_{ic} + \tau_{ia}) \quad (10)$$

where $h(z_i) = M_i^{-1}[\tau_{id} - f_i(\eta_i, v_i)] - \dot{v}_i^d$ is the unknown dynamics, encompassing unknown vehicle dynamics and disturbances, and $g_i = M_i^{-1}$ denotes the input gain matrix.

In scenarios where malicious attackers influence the control input of USVs, it is essential to recognize that DAs are strategic. Consequently, in the Stackelberg game framework, an optimal SSC policy is formulated by considering optimal DA policy to accomplish the target surrounding control of multiple USVs under the targeted DAs. In the context of the Stackelberg game involving the velocity error dynamics (10), the sequential decision-making process unfolds between the attacker DA and the controller SSC, described as follows:

- 1) The sequential decision-making process stems from the initial policy τ_{ic}^0 employed for the controller of USV i .
- 2) Afterward, the attacker, considering the leader's policy, chooses the optimal DA policy τ_{ia}^* as its best response to interacting with the leader based on the current state of the controller, aiming to maximize the errors e_i and z_i .
- 3) In turn, the controller takes on the role of the leader, actively adjusting its strategy by considering the follower's policy to select the optimal SSC policy τ_{ic}^* for minimizing the errors e_i and z_i .

In the sequential decision-making cycle, the leader and the follower autonomously make decisions at distinct levels. Thus, this article establishes the Stackelberg game, wherein the leader's decision is contingent upon the rational choices made by the follower. Both players aim to minimize their cost functions. The performance indexes for the leader J_{ic} and follower J_{ia} are defined as follows:

$$J_{ia}(z_i, \tau_{ia}) = \int_0^\infty r_{ia}(z_i, \tau_{ia}, \tau_{ic})ds \quad (11)$$

$$J_{ic}(z_i, \tau_{ic}) = \int_0^\infty r_{ic}(z_i, \tau_{ia}, \tau_{ic})ds \quad (12)$$

where $r_{ia}(z_i, \tau_{ia}, \tau_{ic}) = \tau_{ia}^\top G_i \tau_{ia} + \tau_{ic}^\top L_i \tau_{ia} - z_i^\top Q_i z_i$ and $r_{ic}(z_i, \tau_{ia}, \tau_{ic}) = z_i^\top Q_i z_i + \tau_{ic}^\top R_i \tau_{ic}$ denote cost functions of the follower and leader with G_i , L_i , Q_i , and R_i denoting symmetric positive definite matrices.

C. Design of Distributed Optimal SSC and DA Policies

In this section, the leader-follower coupled HJ equation for the Stackelberg equilibrium is formulated to obtain the optimal policies for the follower and the leader. The Stackelberg game is delineated as a two-stage optimal control.

Firstly, the attacker acts as the follower, minimizing the performance index J_{ia} based on the leader's information to maximize the disruption of the target surrounding control objective for multiple USVs. Therefore, given the leader control policy of the USV i , the performance index J_{ia} at time t is referred to as the value function given by

$$V_{ia}(z_i(t)) = \int_t^\infty r_{ia}(z_i, \tau_{ia}, \tau_{ic})ds \quad (13)$$

The optimal value function for the follower DA is given by

$$V_{ia}^* = \min_{\tau_{ia}} \int_t^\infty r_{ia}(z_i, \tau_{ia}, \tau_{ic})ds \quad (14)$$

Considering the velocity error dynamics (10), the Hamiltonian function for the follower is defined as

$$H_{ia}(z_i, \tau_{ic}, \tau_{ia}, \nabla V_{ia}) = r_{ia}(z_i, \tau_{ia}, \tau_{ic}) + \nabla V_{ia}^\top [h_i(z_i) + g_i \tau_{ic} + g_i \tau_{ia}] \quad (15)$$

where $\nabla V_{ia} = \partial V_{ia}(z_i(t))/\partial z_i$ represents the partial derivative of $V_{ia}(z_i(t))$ corresponding to z_i . Owing to $\dot{\chi}_{i1} = \dot{V}_{ia} = -\partial H_{ia}(z_i, \tau_{ic}, \tau_{ia}, \nabla V_{ia})/\partial z_i$, the costate equation is $\dot{\chi}_{i1} = 2Q_i z_i - \nabla h_i^\top \nabla V_{ia}$ with $\nabla h_i = \partial h_i(z_i)/\partial z_i$. According to the minimal principle and the fact that $\partial H_{ia}(z_i, \tau_{ic}, \tau_{ia}, \nabla V_{ia}^*)/\partial \tau_{ia} = 0$ with $\nabla V_{ia}^* = \partial V_{ia}^*/\partial z_i$, the optimal DA policy for the follower is derived as

$$\tau_{ia}^* = -\frac{1}{2} G_i^{-1} (g_i^\top \nabla V_{ia}^* + L_i \tau_{ic}) \quad (16)$$

Given the follower's policy, the controller assumes the leader's role in the Stackelberg game, striving to minimize the performance J_{ic} and thereby accomplishing distributed SSC for multiple USVs. The follower selects the decision outlined in (16) as the DA signal and the leader's strategy is additionally influenced by the costate constraints. To this end, the leader's value function of USV i is given as

$$V_{ic}(z_i(t)) = \int_t^\infty [r_{ic}(z_i, \tau_{ia}, \tau_{ic}) + \beta_i^\top \dot{\chi}_{i1}] ds \quad (17)$$

where β_i represents the Lagrangian multiplier. Then, the optimal value function of the leader SSC is denoted as

$$V_{ic}^* = \min_{\tau_{ia}} \int_t^\infty [r_{ic}(z_i, \tau_{ia}, \tau_{ic}) + \beta_i^\top \dot{\chi}_{i1}] ds \quad (18)$$

Consequently, the leader's Hamiltonian function is defined by considering the follower's optimal policy τ_{ia}^* and the costate constraint $\dot{\chi}_{i1}$:

$$H_{ic}(z_i, \tau_{ic}, \tau_{ia}^*, \nabla V_{ic}) = r_{ic}(z_i, \tau_{ia}, \tau_{ic}) + \beta_i \dot{\chi}_{i1} + \nabla V_{ic}^\top [h_i(z_i) + g_i \tau_{ic} + g_i \tau_{ia}^*] \quad (19)$$

where $\nabla V_{ic} = \partial V_{ic}(z_i(t))/\partial z_i$.

Similarly, utilizing the necessary conditions for optimality, the optimal SSC policy τ_{ic}^* and time derivative of β_i are acquired as

$$\tau_{ic}^* = -\frac{1}{2} R_i^{-1} \Psi_i g_i^\top \nabla V_{ic}^* \quad (20)$$

$$\dot{\beta}_i = -\frac{\partial H_{ic}}{\partial (\nabla V_{ia}^*)} = \nabla h_i \beta_i^\top + \frac{1}{2} g_i G_i^{-1} g_i^\top \nabla V_{ic}$$

where $\nabla V_{ic}^* = \partial V_{ic}^*/\partial z_i$, $\Psi_i = I_3 - \frac{1}{2} G_i^{-1} L_i$ represents positive definite matrix. Note that the optimal value functions V_{ia}^* and V_{ic}^* comply with $0 = \min_{\tau_{ia}} H_{ia}(z_i, \tau_{ic}, \tau_{ia}, \nabla V_{ia}^*)$ and $0 = \min_{\tau_{ic}} H_{ic}(z_i, \tau_{ic}, \tau_{ia}^*, \nabla V_{ic}^*)$, respectively.

The leader-follower coupled HJ equation is derived by combining the distributed optimal SSC and DA policies (16) and (20) of the USV i and given as follows:

$$-z_i^\top Q_i z_i + (\tau_{ia}^*)^\top G_i \tau_{ia}^* + \tau_{ic}^\top L_i \tau_{ia}^* + \nabla (V_{ia}^*)^\top [h_i(z_i) + g_i \tau_{ic} + g_i \tau_{ia}^*] = 0$$

$$z_i^\top Q_i z_i + (\tau_{ic}^*)^\top R_i \tau_{ic}^* + \beta_i (2Q_i z_i - \nabla h_i^\top \nabla V_{ia}^*)$$

$$+ (\nabla V_{ic}^*)^\top [h_i(z_i) + g_i \tau_{ic}^* + g_i \tau_{ia}^*] = 0 \quad (21)$$

The stability and Nash equilibrium of the velocity error dynamics with DAs (10) will be analyzed under the optimal SSC policy τ_{ic}^* in (20).

Theorem 1: The velocity error dynamics (10), constrained by the cost function (12) and the optimal DA policy τ_{ia}^* , are guaranteed to be asymptotically stable through the optimal SSC policy τ_{ic}^* . Furthermore, the Stackelberg equilibrium is achieved by the pair $\{\tau_{ic}^*, \tau_{ia}^*(\tau_{ic}^*)\}$ if and only if the leader-follower coupled HJ equations (21) are satisfied by the optimal value functions V_{ia}^* and V_{ic}^* .

Proof: Detailed proof is seen in Appendix A.

Remark 2: The covert and targeted nature of DAs is evident in the attacker's role as a follower. The attacker adjusts its strategy by considering the current SSC policy and tracking errors of the controller, striving to maximize the performance index while minimizing the employment of deception signals. The formulation of this attack strategy benefits from the construction of optimal SSC and DA policies, which adhere to the leader-follower coupled HJ equation within the Stackelberg game framework.

IV. RL-BASED STACKELBERG GAME SOLUTION AND STABILITY ANALYSIS

To achieve available optimal solutions for SSC and DA, computing the gradient of the value function by solving (21) is crucial. However, due to the intricate nonlinearity inherent in the leader-follower coupled HJ equations and the unknown dynamics of USVs, obtaining the analytical solution proves exceedingly challenging or even impossible. Hence, a learning structure comprising two NN-based approximators, termed critic and actor NNs, is employed to approximate the optimal value functions and control policies, respectively. By integrating the formulated Bellman error equation into the actor-critic RL structure, distributed optimal SSC and DA solutions minimizing the cost functions are derived. This approach avoids the difficulty in acquiring the USVs' dynamics. Furthermore, rigorous stability analysis demonstrates the boundedness of all signals in the closed-loop system for multiple USVs.

A. Solution of Follower DA

By employing the NNs-based actor-critic learning structure to approximate the optimal control solution of the follower, it can simultaneously optimize and update both the optimal value function and the DA policy. The optimal value function for the follower is approximated as

$$V_{ia}^* = (W_c^{ia*})^\top \varphi_c^{ia} + \varepsilon_c^{ia} \quad (22)$$

where $W_c^{ia*} \in \mathbb{R}^{M_a}$ denotes the bounded optimal weight of the follower's critic NN, i.e., $\|W_c^{ia*}\| \leq b_{w_c}^{ia}$, M_a is the number of basis neurons, and $\varphi_c^{ia} = \varphi_c^{ia}(z_i)$ and ε_c^{ia} represent the regression function and approximation error, respectively.

Given the unknown nature of the ideal weight W_c^{ia*} , the approximations of V_{ia}^* and ∇V_{ia}^* can be formulated as

$$\hat{V}_{ia} = \hat{W}_c^{ia\top} \varphi_c^{ia}, \nabla \hat{V}_{ia} = \nabla \varphi_c^{ia\top} \hat{W}_c^{ia} \quad (23)$$

where $\hat{W}_c^{ia}$ is the estimation of W_c^{ia*} , $\nabla \hat{V}_{ia} = \partial \hat{V}_{ia} / \partial z_i$, and $\nabla \varphi_c^{ia} = \partial \varphi_c^{ia} / \partial z_i$ is bounded, i.e., $\nabla \varphi_c^{ia} \leq b_{\nabla \varphi_c^{ia}}$.

To avoid using the unknown dynamics $h_i(z_i)$ during the learning process, the Bellman error equation for any time integral h is established as

$$e_c^{ia} = \int_{t-h}^t r_{ia}(z_i, \hat{\tau}_{ia}, \hat{\tau}_{ic}) ds + \hat{W}_c^{ia\top} \Delta \varphi_c^{ia} \quad (24)$$

where $r_{ia}(z_i, \hat{\tau}_{ia}, \hat{\tau}_{ic}) = \hat{\tau}_{ia}^\top G_i \hat{\tau}_{ia} + \hat{\tau}_{ic}^\top L_i \hat{\tau}_{ia} - z_i^\top Q_i z_i$ with $\hat{\tau}_{ia}$ and $\hat{\tau}_{ic}$ being the follower and leader policies, respectively, to be designed later, and $\Delta \varphi_c^{ia} = \varphi_c^{ia}(z_i(t)) - \varphi_c^{ia}(z_i(t-h))$.

To minimize the error e_c^{ia} by adjusting the critic weight $\hat{W}_c^{ia}$, the objective function is defined as $E_{1c} = \frac{1}{2}(e_c^{ia})^2$. Subsequently, the follower's update law of critic NN is formulated using the gradient descent scheme as

$$\dot{\hat{W}}_c^{ia} = -\Gamma_c^{ia} \frac{\Delta \varphi_c^{ia}}{(1 + \Delta \varphi_c^{ia\top} \Delta \varphi_c^{ia})^2} e_c^{ia} \quad (25)$$

where Γ_c^{ia} denotes the learning rate of follower's critic NN.

An actor NN is introduced to ensure the stability of policy and value function updates, instead of employing a single update law in [35]. Specifically, the actor NN for the follower DA is designed as

$$\hat{\tau}_{ia} = -\frac{1}{2} G_i^{-1} (g_i^\top \nabla \varphi_c^{ia\top} \hat{W}_a^{ia} + L_i \hat{\tau}_{ic}) \quad (26)$$

where $\hat{W}_a^{ia}$ denotes the actor weight for estimating the weight W_a^{ia*} and the update law for the actor NN can be derived as

$$\begin{aligned} \dot{\hat{W}}_a^{ia} = & -\Gamma_a^{ia} \left[k_a^{ia} \hat{W}_a^{ia} - \frac{1}{2} \Pi_c^{ia} z_i \right. \\ & \left. - \frac{h}{4(1 + \Delta \varphi_c^{ia\top} \Delta \varphi_c^{ia})^2} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \hat{W}_a^{ia} \Delta \varphi_c^{ia\top} \hat{W}_c^{ia} \right] \end{aligned} \quad (27)$$

with Γ_a^{ia} denoting the learning rate of the follower's actor NN and k_a^{ia} being a positive constant. Obviously, $\Pi_c^{ia} = \nabla \varphi_c^{ia} g_i G_i^{-1} g_i^\top$ is bounded, i.e., $\Pi_c^{ia} \leq b_{\Pi_c^{ia}}$.

B. Solution of Leader SSC

This section derives the leader DA's optimal control solution, similar to the follower's approach. The distinction lies in integrating the value function decomposition (VFD) technique, which fine-tunes the leader's optimal SSC policy by leveraging parameter information from the controller. For the unknown dynamics $h_i(z_i)$ that is inaccessible in items $\dot{\chi}_{i1}$ and β_i , the additional NN is utilized to estimate the dynamics as

$$h_i(z_i) = (W_{ih}^*)^\top \varphi_{ih} + \varepsilon_{ih}^* \quad (28)$$

where $W_{ih}^* = [W_{ih}^{u*}, W_{ih}^{v*}, W_{ih}^{r*}] \in R^{M_h \times 3}$ is the bounded optimal weight, i.e., $\|W_{ih}^*\|_F \leq b_{W_{ih}^*}$, M_h is the number of hidden layer neurons, and $\varphi_{ih} = \varphi_{ih}(z_i)$ and ε_{ih} denote the regression

function and approximation error, respectively, both of which are bounded, i.e., $\|\varphi_{ih}\| \leq b_{\varphi_{ih}}^{ih}$ and $\|\varepsilon_{ih}\| \leq b_{\varepsilon_{ih}}^{ih}$. Likewise, the approximation of the unknown dynamics and its gradients is expressed as

$$\hat{h}_i(z_i) = \hat{W}_{ih}^\top \varphi_{ih}, \nabla \hat{h}_i = \nabla \varphi_{ih}^\top \hat{W}_{ih} \quad (29)$$

with $\hat{W}_{ih}$ denoting the estimation of W_{ih}^* , $\nabla \hat{h}_i = \partial \hat{h}_i / \partial z_i$, and $\nabla \varphi_{ih} = \partial \varphi_{ih} / \partial z_i$.

To expedite the solution of optimal policies and enhance the safety of the target surrounding control for multiple USVs, without compromising system stability, the optimal value function is decomposed by fully utilizing the parameters designed by the controller:

$$V_{ic}^* = (z_i^\top K_{zi} + e_i^\top \Re(\eta_i) + \hat{h}_i^\top(z_i)) \Pi_{ic} z_i + V_{ic}^0 \quad (30)$$

where $V_{ic}^0 = V_{ic}^* - V_{ic}^1$ with $V_{ic}^1 = (z_i^\top K_{zi} + e_i^\top \Re(\eta_i) + \hat{h}_i^\top(z_i)) z_i$, $\Pi_{ic} = 2(g_i^\top)^{-1} R_i g_i^\top (\Psi_i)^{-1}$, and K_{zi} is a positive definite matrix. By substituting (30) into (20), the optimal policy for leader SSC is reconstructed as

$$\begin{aligned} \tau_{ic}^* = & -g_i^{-1} (K_{zi} z_i + \Re(\eta_i)^\top e_i + \hat{W}_{ih}^\top \varphi_{ih}) \\ & - \frac{1}{2} R_i^{-1} \Psi_i g_i^\top \nabla V_{ic}^0 \end{aligned} \quad (31)$$

with $\nabla V_{ic}^0 = \partial V_{ic}^0 / \partial z_i$.

The leader's optimal value function is approximated as

$$V_{ia}^* = (W_c^{ic*})^\top \varphi_c^{ic} + \varepsilon_{ic}^{ic} + V_{ic}^1 \quad (32)$$

where $W_c^{ic*} \in R^{M_c}$ denotes the optimal weight of critic NN and is bounded, i.e., $\|W_c^{ic*}\| \leq b_{W_c^{ic*}}^{ic}$, M_c is the number of basis regressor, $\varphi_c^{ic} = \varphi_c^{ic}(z_i)$ and ε_{ic}^{ic} denote the regression function and approximation error, respectively.

Similarly, the approximation of the optimal value function, referred to as the leader's critic NN, can be defined as

$$\begin{aligned} \hat{V}_{ic} = & \hat{V}_{ic}^0 + V_{ic}^1 = \hat{W}_c^{ic\top} \varphi_c^{ic} + V_{ic}^1 \\ \nabla \hat{V}_{ic} = & \nabla \hat{V}_{ic}^0 + \nabla V_{ic}^1 = \nabla \varphi_c^{ic\top} \hat{W}_c^{ic} + \nabla V_{ic}^1 \end{aligned} \quad (33)$$

where $\hat{W}_c^{ic}$ denotes the estimation of W_c^{ic*} , $\nabla \hat{V}_{ic} = \partial \hat{V}_{ic} / \partial z_i$, $\nabla \hat{V}_{ic}^0 = \partial \hat{V}_{ic}^0 / \partial z_i$, $\nabla V_{ic}^1 = \partial V_{ic}^1 / \partial z_i$, and $\nabla \varphi_c^{ic} = \partial \varphi_c^{ic} / \partial z_i$ is bounded, i.e., $\nabla \varphi_c^{ic} \leq b_{\nabla \varphi_c^{ic}}$. Then, the Bellman error equation and the updated law of the leader's critic weight are given as

$$\begin{aligned} e_c^{ic} = & \int_{t-h}^t [r_{ic}(z_i, \hat{\tau}_{ia}, \hat{\tau}_{ic}) + \beta_i \dot{\chi}_{i1}] ds + \hat{W}_c^{ic\top} \Delta \varphi_c^{ic} + \Delta V_{ic}^1 \\ \dot{\hat{W}}_c^{ic} = & -\Gamma_c^{ic} \frac{\Delta \varphi_c^{ic}}{(1 + \Delta \varphi_c^{ic\top} \Delta \varphi_c^{ic})^2} e_c^{ic} \end{aligned} \quad (34)$$

with $r_{ic}(z_i, \hat{\tau}_{ia}, \hat{\tau}_{ic}) = z_i^\top Q_i z_i + \hat{\tau}_{ic}^\top R_i \hat{\tau}_{ic}$, $\dot{\chi}_{i1} = 2Q_i z_i - \hat{W}_{ih}^\top \nabla \varphi_{ih} \nabla \varphi_c^{ia\top} \hat{W}_a^{ia}$, $\Delta \varphi_c^{ic} = \varphi_c^{ic}(z_i(t)) - \varphi_c^{ic}(z_i(t-h))$, and $\Delta V_{ic}^1 = V_{ic}^1(t) - V_{ic}^1(t-h)$.

Thus, considering the above analysis, the approximation of the optimal SSC policy for the USV i is formulated as

$$\begin{aligned} \hat{\tau}_{ic} = & -g_i^{-1} (K_{zi} z_i + \Re(\eta_i)^\top e_i + \hat{W}_{ih}^\top \varphi_{ih}) \\ & - \frac{1}{2} R_i^{-1} \Psi_i g_i^\top \nabla \varphi_c^{ic\top} \hat{W}_a^{ic} \end{aligned} \quad (35)$$

where $\hat{W}_a^{ic}$ is the leader's actor weight and it is updated by

$$\begin{aligned} \dot{\hat{W}}_a^{ic} = & -\Gamma_a^{ic}[k_a^{ic}\hat{W}_a^{ic} - \frac{1}{2}\Pi_c^{ic}z_i \\ & - \frac{h}{4(1 + \Delta\varphi_c^{ic\top}\Delta\varphi_c^{ic})^2}\Pi_c^{ic}\nabla\varphi_c^{ic\top}\hat{W}_a^{ic}\Delta\varphi_c^{ic\top}\hat{W}_c^{ic}] \end{aligned} \quad (36)$$

with Γ_c^{ic} denoting the learning rate of the leader's actor NN, k_a^{ic} being a positive constant. Clearly, $\Pi_c^{ic} = \nabla\varphi_c^{ic}g_i\Psi_i^\top R_i^{-1}\Psi_i g_i^\top$ is bounded, i.e. $\Pi_c^{ic} \leq b_{\pi_c}^{ic}$.

For the additional NN, it can approximate an unknown dynamics $h_i(z_i)$ by minimizing the error z_i . To achieve this, the update law of the additional NN is designed as

$$\dot{\hat{W}}_{ih}^\top = -k_{ih}\Gamma_{ih}\hat{W}_{ih} + \Gamma_{ih}\varphi_{ih}z_i^\top \quad (37)$$

where Γ_{ih} and k_{ih} denote the learning rate matrix and positive constant, respectively.

C. Stability Analysis

Lemma 1: For system (7) with the state vector Y_t and input vector $\tilde{\eta}_t$, the existence of $\Theta_t = P_t^{-1}C^\top$ with λ_{it} and $a > \max_{i=1,\dots,N}\{1/\lambda_{it}\}$ denoting the i th eigenvalue of H_t ensures that the system is input-to-state stable. Additionally, under Assumption 1, the estimated error vector $\tilde{\eta}_{it}$ is ultimately uniformly bounded (UUB).

Proof: Considering the candidate Lyapunov function as $\mathcal{V}_t = Y_t^\top(I_N \otimes P_t)Y_t/2$, the time derivative along with (7) is denoted by $\dot{\mathcal{V}}_t = Y_t^\top(P_t A + A^\top P_t - 2cH_t \otimes P_t \Theta_t C)Y_t/2 - Y_t^\top(1_N \otimes (B\tilde{\eta}_t))$. Given the connected nature of graph $\mathcal{G}$ and the symmetry of the matrix H_t , it is established that M_t is positive definite. Consequently, there exists a unitary matrix $U_t \in R^{N \times N}$ such that $U_t^\top H_t U_t = \Lambda_t \triangleq \text{diag}\{\lambda_{1t}, \dots, \lambda_{Nt}\}$. Through the application of $\bar{Y}_t = (U_t^\top \otimes I_6)Y_t$, $\Theta_t = P_t^{-1}C^\top$, and $c > \max_{i=1,\dots,N}\{1/\lambda_{it}\}$, it yields that $\dot{\mathcal{V}}_t = \bar{Y}_t^\top(I_N \otimes (P_t A + A^\top P_t) - 2c\Lambda_t \otimes C^\top C)\bar{Y}_t/2 - Y_t^\top(1_N \otimes (B\tilde{\eta}_t)) \leq -\lambda_{\min}(Q)\|\bar{Y}_t\|^2 + \sqrt{N}\|P_t B\|\|\tilde{\eta}_t\|\|\bar{Y}_t\|$. Given $\|\bar{Y}_t\| \geq 2\sqrt{N}\|P_t B\|\|\tilde{\eta}_t\|/(\theta_t \lambda_{\min}(Q))$, the deduction $\dot{\mathcal{V}}_t \leq -(1 - \theta_t)\lambda_{\min}(Q)\|\bar{Y}_t\|^2$ implies that the subsystem (7) is input-to-state stable, and

$$\|\bar{Y}_t\| \leq \max\{\varpi_t(\|\bar{Y}_t\|, t), \phi_{\tilde{\eta}_t}(\|\tilde{\eta}_t\|)\} \quad (38)$$

where ϖ_t denotes a class $\mathcal{KL}$ function and $\phi_{\tilde{\eta}_t}(x) = \sqrt{N}\lambda_{\max}(I_N \otimes P_t)/\lambda_{\min}(I_N \otimes P_t)\|P_t B\|x/(\theta_t \|\bar{Y}_t\|)$.

By Assumption 1 and (1), it can be inferred that $\|\tilde{\eta}_t\|$ is bounded, resulting in $\|\bar{Y}_t\|$ remaining UUB. Furthermore, taking into account of $\bar{Y}_t = (U_t^\top \otimes I_6)Y_t$, the estimation error $\tilde{\eta}_{it}$ is also UUB. This completes the proof.

Theorem 2: Consider the kinematics and dynamics of USVs (2) and the target kinematics (1) under Assumptions 1 and 2, the distributed SSC policy with the update laws (34), (36), and (37) under the optimal DA policy with the update laws (25) and (27) can ensure that all errors in the closed-loop system of multiple USVs, given bounded initial conditions, are UUB.

Proof: Detailed proof is seen in Appendix B.

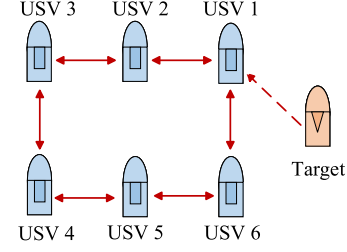


Fig. 3. Interaction topology graph.

V. SIMULATION RESULTS

In this section, extensive simulation results are presented to validate the effectiveness and security of the proposed distributed SSC policy under DAs. The simulation scenario involves six USVs and one target agent, with their interaction topology illustrated in Fig. 3. The model parameters for the USVs are specified based on CyberShip II, as detailed in [32]. The external disturbances affecting USVs are characterized as first-order Gauss–Markov processes, denoted by $\dot{\tau}_{id} = w_i - 0.5\tau_{id}$, where $w_i \in \mathbb{R}^3$ denotes a white noise. The position/heading vector of the USVs is initialized as $\eta_1(0) = [-1, 7, 0]^\top$, $\eta_2(0) = [-5, 7, 0]^\top$, $\eta_3(0) = [-8, 2, 0]^\top$, $\eta_4(0) = [-5, -7, 0]^\top$, $\eta_5(0) = [-1, -7, 0]^\top$, $\eta_6(0) = [4, -2, 0]^\top$. The position/heading information of the target agent is exclusively known by USV 1, with the initial vector of the target set as $\eta_t(0) = [0, 0, 0]^\top$. The target velocity is taken as $v_t = [0.3, 0.05, 0.02 \cos(0.02t)]^\top$, and it is unavailable for any USVs. The attacker's policy (26) serves as an attack signal to disrupt the collaboration among multiple USVs, while the controller's policy (35) is implemented to guide the USVs in resisting attacks and enclosing the target with the predefined radius $\rho = 6$ m. Furthermore, the parameters for the distributed target estimator are chosen as $c = 2$, $\vartheta_{t1} = I_3$, $\vartheta_{t2} = 2I_3$, the parameters for the follower DA are specified as $G_i = 0.05I_3$, $L_i = 0.05I_3$, $Q_i = 20I_3$, $\Gamma_c^{ia} = 0.9I_3$, $\Gamma_a^{ia} = 0.7I_3$, $k_a^{ia} = 0.6$ and the parameters for the leader SSC are set as $K_{ei} = I_3$, $K_{zi} = 1.5I_3$, $\Gamma_c^{ic} = 0.8I_3$, $\Gamma_a^{ic} = 0.7I_3$, $\Gamma_{ih} = 0.8I_3$, $k_a^{ic} = 0.6$, $k_{ih} = 0.6$.

The simulation results under the attacker's policy (26) are given in Figs. 4–8. The target surrounding trajectory involving one target and six USVs is displayed in Fig. 4. It demonstrates that the proposed distributed SSC policy allows USVs to surround the target in a uniformly spaced regular hexagonal formation while consistently aligning with the target's heading angle, even under more covert and targeted DAs. The control objective $\bar{e}_{it} = \eta_i - \eta_t - p_{it} = [\bar{x}_e^i, \bar{y}_e^i, \bar{\psi}_e^i]^\top$ for multiple USVs utilizing the proposed control scheme is illustrated in Fig. 5. It indicates that even in scenarios where global target pose information is inaccessible, the control error manages to converge to a confined neighborhood of the origin under the follower strategy attacks. The outcomes depicted in Fig. 6 reveal that the tracking errors e_i and z_i and the approximation error $\tilde{h}_i(z_i)$ are confined within bounds. It suggests that by tracking the intermediate control law, the target surrounding error e_i can be minimized, ultimately facilitating the achievement of distributed SSC for multiple USVs. Furthermore, the additional NN is proved to be

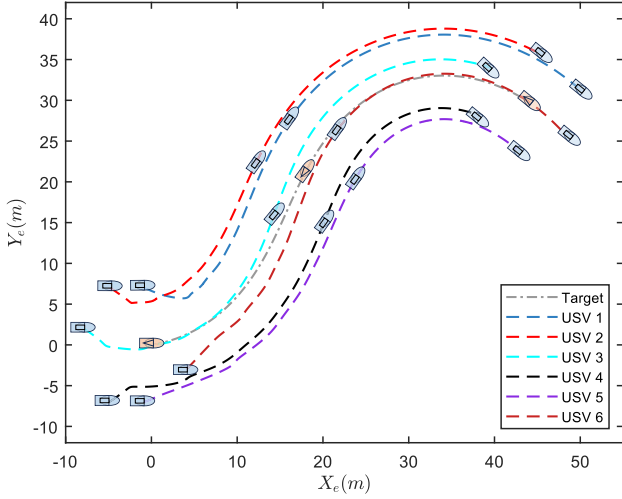
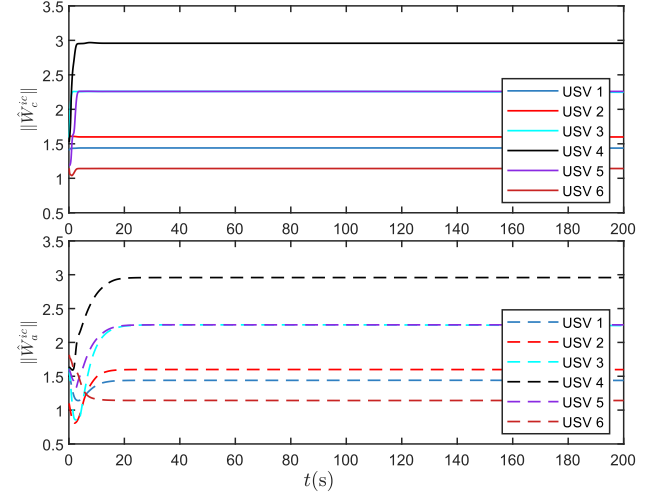
Fig. 4. Target surrounding process for multiple USVs under the attacks $\hat{\tau}_{ia}$.

Fig. 7. Convergence of actor-critic NNs for the leader SSC policy.

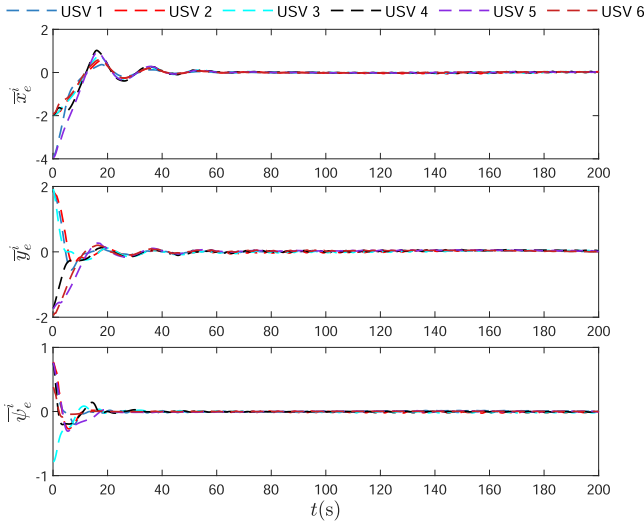
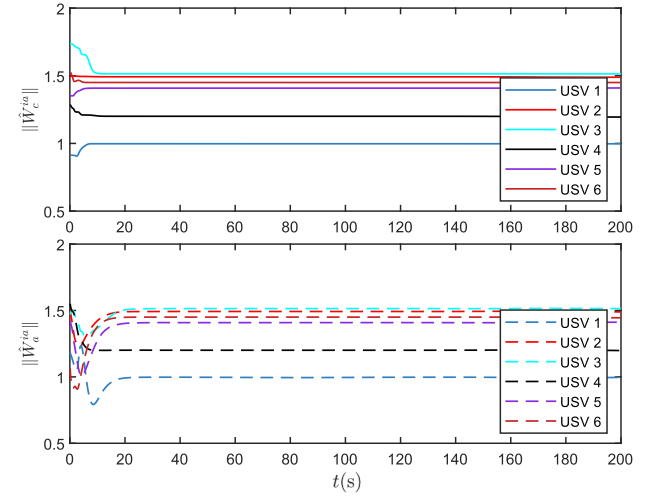
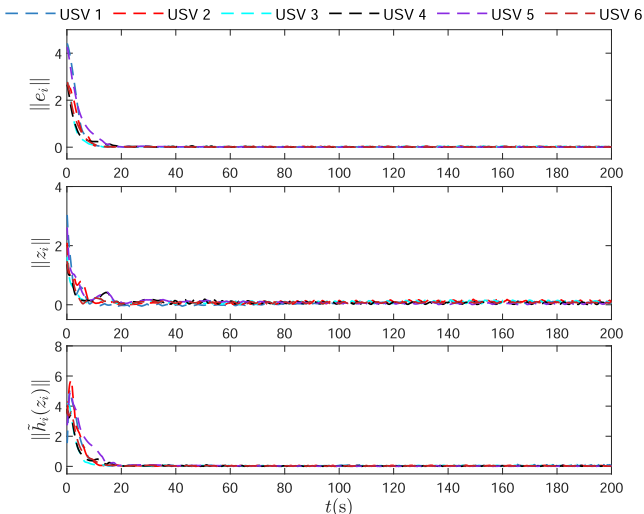
Fig. 5. Control objective of target surrounding under the attacks $\hat{\tau}_{ia}$.

Fig. 8. Convergence of actor-critic NNs for the follower DA policy.

Fig. 6. Errors e_i , z_i , and $\tilde{h}_i(z_i)$ of target surrounding under the attacks $\hat{\tau}_{ia}$.

effective in approximating the unknown dynamics. Figs. 7 and 8 depict the convergence of the actor-critic NNs for both the leader and follower. It is shown that the actor-critic NNs can rapidly converge with the assistance of additional NNs. Thus, the optimal distributed SSC and DA policies can be effectively approximated.

To further validate the effectiveness of the distributed optimal SSC policy acquired through online learning with the proposed control scheme, the deception signals are treated as bounded functions: $\tau_{ia} = [19 + 9 \sin(0.1t) \cos(0.5t), 22 + 6 \sin(0.5t), 6 + 2 \cos(0.5t)]^\top$. The deception signals are injected into USV 2, USV 3, and USV 4 at time instants $t \in [60 \text{ s}, 67 \text{ s}]$, $t \in [85 \text{ s}, 95 \text{ s}]$, and $t \in [120 \text{ s}, 135 \text{ s}]$, respectively. The control objective $\bar{e}_{it}$ for multiple USVs utilizing the optimal SSC policy is displayed under the deception signals in Fig. 9. It reveals that the optimal SSC policy derived from the proposed scheme maintains effective cooperative surrounding control performance even when deception signals attack the USVs at any moment.

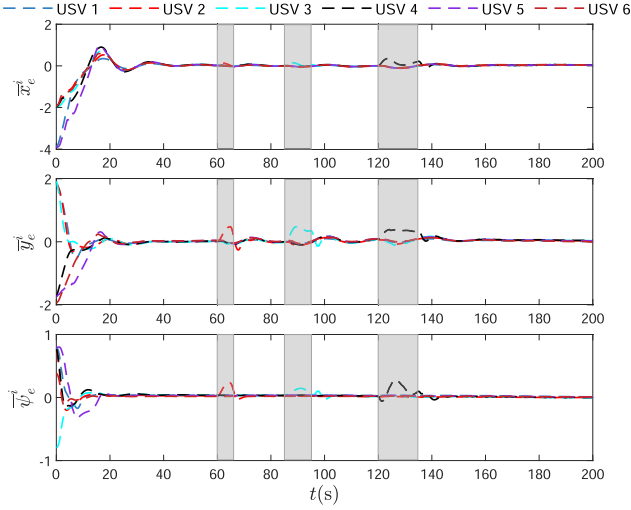


Fig. 9. Control objective of target surrounding under DAs with bounded functions.

TABLE I
CONTROL PERFORMANCE WITH DIFFERENT METHODS

Method	Conv time(s)	IAE		
		$\ \bar{e}\ $	$\ e\ $	$\ z\ $
the proposed scheme	18.9812	83.6565	34.2589	63.2460
the proposed scheme without VFD	27.3353	113.9843	54.2379	79.8341

Furthermore, a simulation comparison is conducted on the proposed control scheme with and without the VFD technique in the presence of strategic attack signals (26). The following indicators are employed to delineate distinctions between the two methods: Conv time (elapsed time from the initiation point to convergence for actor-critic NNs) and IAE (integral absolute error). The error vectors are denoted as $\bar{e} = [\bar{e}_1, \dots, \bar{e}_6]^T$, $e = [e_1, \dots, e_6]^T$, and $z_t = [z_1, \dots, z_6]^T$. Table I presents the Conv time and IAE of the error vector under two distinct control methods. It can be indicated that the proposed scheme with VFD exhibits a faster convergence velocity of actor-critic NNs and smaller errors under the attacker's policy (26).

VI. CONCLUSION

In this article, a novel Stackelberg game-based distributed SSC scheme for multiple USVs is presented to operate amidst covert and targeted DAs while encircling a moving target. The distributed target estimator is proposed for pose and velocity estimation. Through the estimator and an intermediate control law, the target surrounding scenario is transformed into the Stackelberg game-solving issue. The NNs-based actor-critic learning structure, complemented by an additional NN, is designed to derive the optimal SSC and DA policies without prior knowledge of vehicle dynamics. Furthermore, the optimal SSC policy, incorporating the VFD technique, effectively expedites the acquisition of optimal policies, mitigating the impact of DAs. The stability of the closed-loop system is verified through

Lyapunov analysis, while simulation results underscore the effectiveness and safety of the proposed scheme against DAs for multiple USVs. Future studies could significantly delve into the surrounding control problem of multiple targets for multiple USVs under complex actuator faults in the physical hierarchy and sensor attacks in the networked hierarchy, further leveraging the improved Stackelberg game-based distributed SSC scheme.

APPENDIX A PROOF OF THEOREM 1

Part I: To prove the stability of the tracking error dynamics, the function $\dot{V}_{ic}^*$ is chosen as a Lyapunov function candidate. Its derivative with (21) is computed as

$$\begin{aligned}\dot{V}_{ic}^* &= \left(\frac{\partial V_{ic}^*}{\partial z_i} \right) \dot{z}_i + \left(\frac{\partial V_{ic}^*}{\partial \nabla V_{ia}^*} \right) \dot{V}_{ia}^* \\ &= -z_i^T Q_i z_i - (\tau_{ic}^*)^T R_i \tau_{ic}^*\end{aligned}$$

Since $Q_i > 0$ and $R_i > 0$, it can be concluded that $\dot{V}_{ic}^* < 0$, meaning that the tracking error system (10) is asymptotically stable when the optimal policy τ_{ic}^* is employed.

Part II: To prove Stackelberg Equilibrium, the performance index (11) is reformulated as

$$\begin{aligned}J_{ia}(z_i(0), \tau_{ia}, \tau_{ic}) &= \int_0^\infty (-z_i^T Q_i z_i + \tau_{ia}^T G_i \tau_{ia} + \tau_{ic}^T L_i \tau_{ic}) ds \\ &\quad + \int_0^\infty \dot{V}_{ia}^* ds + V_{ia}^*(z_i(0)) - V_{ia}^*(z_i(\infty))\end{aligned}$$

By combining $\dot{V}_{ia}^* = (\nabla V_{ia}^*)^T [h(z_i) + g_i \tau_{ic} + g_i \tau_{ia}]$ and $-2(\tau_{ia}^*)^T G_i \tau_{ia} = (\nabla V_{ia}^*)^T g_i \tau_{ia} + \tau_{ic}^T L_i \tau_{ic}$, where τ_{ia}^* is given by (16), and utilizing the completion of the square method, it yields that

$$\begin{aligned}J_{ia}(z_i(0), \tau_{ia}, \tau_{ic}) &= \int_t^\infty (\tau_{ia} - \tau_{ia}^*)^T G_i (\tau_{ia} - \tau_{ia}^*) ds \\ &\quad + \int_t^\infty (-z_i^T Q_i z_i - (\tau_{ia}^*)^T G_i \tau_{ia}^* + (\nabla V_{ia}^*)^T [h(z_i) + g_i \tau_{ic}]) ds \\ &\quad + V_{ia}^*(z_i(0)) - V_{ia}^*(z_i(\infty))\end{aligned}$$

In view of $-(\tau_{ia}^*)^T G_i \tau_{ia}^* = (\tau_{ia}^*)^T G_i \tau_{ia}^* + (\nabla V_{ia}^*)^T g_i \tau_{ia}^* + \tau_{ic}^T L_i \tau_{ic}^*$, and by employing the leader coupled HJ equation given in (21), it can further be simplified as

$$\begin{aligned}J_{ia}(z_i(0), \tau_{ia}) &= \int_t^\infty (\tau_{ia} - \tau_{ia}^*)^T G_i (\tau_{ia} - \tau_{ia}^*) ds \\ &\quad + V_{ia}^*(z_i(0)) - V_{ia}^*(z_i(\infty))\end{aligned}$$

By setting $\tau_{ia} = \tau_{ia}^*$, the optimal value function for the follower is $J_{ia}(z_i(0), \tau_{ia}^*) = V_{ia}^*(z_i(0)) - V_{ia}^*(z_i(\infty))$. Since $\dot{V}_{ia}^* = z_i^T Q_i z_i - (\tau_{ia}^*)^T G_i \tau_{ia}^* - \tau_{ic}^T L_i \tau_{ic}^*$, when the inequality $z_i^T Q_i z_i \leq \tau_{ic}^T L_i \tau_{ic}^*$ holds, it follows that $J_{ia}(\tau_{ic}, \tau_{ia}^*) \leq J_{ia}(\tau_{ic}, \tau_{ia})$. The same procedure can be performed to demonstrate that $J_{ic}(\tau_{ic}^*, \tau_{ia}^*) \leq J_{ic}(\tau_{ic}, \tau_{ia}^*)$.

As a result, given that the aforementioned conditions are satisfied for the follower DA and leader SSC in the Stackelberg game, the Nash equilibrium $\{\tau_{ic}^*, \tau_{ia}^*(\tau_{ic}^*)\}$ stated in Definition 2 is realized.

APPENDIX B PROOF OF THEOREM 2

Consider the following Lyapunov function candidate

$$\mathcal{V}_i = \mathcal{V}_i^1 + \mathcal{V}_i^2 + \mathcal{V}_i^3 \quad (39)$$

where $\mathcal{V}_i^1 = e_i^\top e_i/2 + z_i^\top z_i/2 + \text{tr}(\tilde{W}_{ih}^\top \Gamma_{ih}^{-1} \tilde{W}_{ih})/2$, $\mathcal{V}_i^2 = \tilde{W}_a^{ic\top} \Gamma_a^{ic-1} \tilde{W}_a^{ic}/2 + \tilde{W}_c^{ic\top} \Gamma_c^{ic-1} \tilde{W}_c^{ic}/2$, and $\mathcal{V}_i^3 = \tilde{W}_c^{ia\top} \Gamma_c^{ia-1} \tilde{W}_c^{ia}/2 + \tilde{W}_a^{ia\top} \Gamma_a^{ia-1} \tilde{W}_a^{ia}/2$, with $\tilde{W}_{ih} = W_{ih}^* - \hat{W}_{ih}$, $\tilde{W}_a^{ic} = W_a^{ic*} - \hat{W}_a^{ic}$, $\tilde{W}_c^{ic} = W_c^{ic*} - \hat{W}_c^{ic}$, $\tilde{W}_a^{ia} = W_a^{ia*} - \hat{W}_a^{ia}$, and $\tilde{W}_c^{ia} = W_c^{ia*} - \hat{W}_c^{ia}$.

Step 1: Taking the derivative of $\mathcal{V}_i^1$ leads to $\dot{\mathcal{V}}_i^1 = e_i^\top \dot{e}_i + z_i^\top \dot{z}_i - \text{tr}(\tilde{W}_{ih}^\top \Gamma_{ih}^{-1} \dot{\tilde{W}}_{ih})$. By applying (9), the first term is $e_i^\top \dot{e}_i = e_i^\top [\Re(\eta_i)(v_i^d + z_i) - \Lambda_i \hat{z}_{it}] = -e_i^\top K_{ei} e_i + e_i^\top \Re(\eta_i) z_i$. Based on (26), (35) and (28), the term $z_i^\top \dot{z}_i$ is given as

$$\begin{aligned} z_i^\top \dot{z}_i &= z_i^\top [h_i(z_i) - K_{zi} z_i - \Re(\eta_i)^\top e_i - \hat{W}_{ih}^\top \varphi_{ih} \\ &\quad - \frac{1}{2} \Pi_c^{ia\top} \hat{W}_a^{ia} - \frac{1}{2} \Pi_c^{ic\top} \hat{W}_a^{ic}] \\ &\leq -z_i^\top K_{zi} z_i + \frac{1}{2} z_i^\top \Pi_c^{ia\top} \tilde{W}_a^{ia} + \frac{1}{2} z_i^\top \Pi_c^{ic\top} \tilde{W}_a^{ic} - z_i^\top \Re(\eta_i)^\top e_i \\ &\quad - z_i^\top \tilde{W}_{ih}^\top \varphi_{ih} + [b_\varepsilon^{ih} + \frac{1}{2} (b_{\pi_c}^{ia} b_{w_c}^{ia} + b_{\pi_c}^{ic} b_{w_c}^{ic})] \|z_i\| \end{aligned}$$

where $b_{z_i} = b_\varepsilon^{ih} + \frac{1}{2} (b_{\pi_c}^{ia} b_{w_c}^{ia} + b_{\pi_c}^{ic} b_{w_c}^{ic})$.

By using (37), the third term in $\dot{\mathcal{V}}_i^1$ is derived as $-\text{tr}(\tilde{W}_{ih}^\top \Gamma_{ih}^{-1} \dot{\tilde{W}}_{ih}) \leq -k_{ih} \text{tr}(\tilde{W}_{ih}^\top \tilde{W}_{ih}) + \text{tr}(\tilde{W}_{ih} \varphi_{ih} z_i^\top) + k_{ih} b_{w_c}^{ih} \|\tilde{W}_{ih}\|_F$. Based on the above derivation, $\dot{\mathcal{V}}_i^1$ can ultimately be rewritten as

$$\begin{aligned} \dot{\mathcal{V}}_i^1 &\leq -e_i^\top K_{ei} e_i - z_i^\top K_{zi} z_i - k_{ih} \text{tr}(\tilde{W}_{ih}^\top \tilde{W}_{ih}) + b_{z_i} \|z_i\| \\ &\quad + k_{ih} b_{w_c}^{ih} \|\tilde{W}_{ih}\|_F + \frac{1}{2} z_i^\top \Pi_c^{ia\top} \tilde{W}_a^{ia} + \frac{1}{2} z_i^\top \Pi_c^{ic\top} \tilde{W}_a^{ic} \end{aligned} \quad (40)$$

Step 2: Differentiating $\mathcal{V}_i^2$ yields the following expression $\dot{\mathcal{V}}_i^2 = -\tilde{W}_a^{ic\top} \Gamma_a^{ic-1} \dot{\tilde{W}}_a^{ic} - \tilde{W}_c^{ic\top} \Gamma_c^{ic-1} \dot{\tilde{W}}_c^{ic}$. Subsequently, substituting (34) and (36) into $\dot{\mathcal{V}}_i^2$, it can be deduced as

$$\begin{aligned} \dot{\mathcal{V}}_i^2 &\leq -k_a^{ic} \tilde{W}_a^{ic\top} \tilde{W}_a^{ic} - k_c^{ic} \tilde{W}_c^{ic\top} \tilde{W}_c^{ic} - \frac{1}{2} \tilde{W}_a^{ic\top} \Pi_c^{ic} z_i \\ &\quad - \frac{h}{4m_{ic}} \tilde{W}_a^{ic\top} \Pi_c^{ic} \nabla \varphi_c^{ic\top} \hat{W}_a^{ic} \Delta \varphi_c^{ic\top} \hat{W}_c^{ic} + k_a^{ic} (W_c^{ic*})^\top \tilde{W}_a^{ic} \\ &\quad + k_c^{ic} (W_c^{ic*})^\top \tilde{W}_c^{ic} + \frac{1}{m_{ic}} \tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} (p_{ic} + \Delta V_{ic}^1) \end{aligned}$$

where $m_{ic} = (1 + \Delta \varphi_c^{ic\top} \Delta \varphi_c^{ic})^2$, $k_c^{ic} = \|\Delta \varphi_c^{ic}\|^2/m_{ic}$, and $p_{ic} = \int_{t-h}^t (r_{ic}(z_i, \hat{\tau}_{ia}, \hat{\tau}_{ic}) + \beta_i \dot{\chi}_{1i}) ds$. And then, combining the expression for $\varepsilon_{hjb}^{ic} = z_i^\top Q_i z_i + (\tau_{ic}^*)^\top R_i \tau_{ic}^* + \beta_i \dot{\chi}_{1i} + (\nabla \varphi_c^{ic\top} W_c^{ic*} + \nabla V_{ic}^1)^\top [h(z_i) + g_i \tau_{ic}^* + g_i \tau_{ia}^*]$, the term $\tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} (p_{ic} + \Delta V_{ic}^1)/m_{ic}$ can be given as

$$\frac{1}{m_{ic}} \tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} (p_{ic} + \Delta V_{ic}^1)$$

$$\begin{aligned} &= \frac{1}{m_{ic}} \tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} [\bar{h}_{ic} - \int_{t-h}^t \nabla V_{ic}^1 d(z_i) + \Delta V_{ic}^1 \\ &\quad + \int_{t-h}^t [\hat{\tau}_{ic}^\top R_i \hat{\tau}_{ic} - (\tau_{ic}^*)^\top R_i \tau_{ic}^*] ds] \\ &= \frac{\bar{h}_{ic}}{m_{ic}} \Delta \varphi_c^{ic\top} \tilde{W}_c^{ic} - \frac{h}{2m_{ic}} \tilde{W}_c^{ic\top} D_c^{ic} \tilde{W}_a^{ic} \\ &\quad + \frac{h}{4m_{ic}} \tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} \tilde{W}_a^{ic\top} \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \end{aligned}$$

with $\bar{h}_{ic} = \int_{t-h}^t \varepsilon_{hjb}^{ic} - (W_c^{ic*})^\top \nabla \varphi_c^{ic} [h_i(z_i) + g_i \tau_{ic}^* + g_i \tau_{ia}^*] ds$, $D_c^{ic} = \Delta \varphi_c^{ic} (\nabla V_{ic}^*)^\top g_i \Psi_i^\top R_i^{-1} \Psi_i g_i^\top \nabla \varphi_c^{ic\top}$ and

$$\begin{aligned} &\frac{h}{4m_{ic}} \tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} \tilde{W}_a^{ic\top} \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \\ &= -\frac{h}{4m_{ic}} (W_c^{ic*})^\top \Delta \varphi_c^{ic} (W_c^{ic*})^\top \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \\ &\quad + \frac{h}{4m_{ic}} \tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} (W_c^{ic*})^\top \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \\ &\quad + \frac{h}{4m_{ic}} (W_c^{ic*})^\top \Delta \varphi_c^{ic} \tilde{W}_a^{ic\top} \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \\ &\quad + \frac{h}{4m_{ic}} \hat{W}_c^{ic\top} \Delta \varphi_c^{ic} \tilde{W}_a^{ic\top} \Pi_c^{ic} \nabla \varphi_c^{ic\top} \hat{W}_a^{ic} \end{aligned}$$

Using the formula derived above, $\dot{\mathcal{V}}_i^2$ can be further reconstructed as

$$\begin{aligned} \dot{\mathcal{V}}_i^2 &\leq -k_a^{ic} \tilde{W}_a^{ic\top} \tilde{W}_a^{ic} + \frac{h}{4m_{ic}} (W_c^{ic*})^\top \Delta \varphi_c^{ic} \tilde{W}_a^{ic\top} \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \\ &\quad - k_c^{ic} \tilde{W}_c^{ic\top} \tilde{W}_c^{ic} + \frac{h}{4m_{ic}} \tilde{W}_c^{ic\top} \Delta \varphi_c^{ic} (W_c^{ic*})^\top \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \\ &\quad - \frac{h}{2m_{ic}} \tilde{W}_c^{ic\top} D_c^{ic} \tilde{W}_a^{ic} + k_a^{ic} (W_c^{ic*})^\top \tilde{W}_a^{ic} + \frac{\bar{h}_{ic}}{m_{ic}} \Delta \varphi_c^{ic\top} \tilde{W}_c^{ic} \\ &\quad - \frac{h}{4m_{ic}} (W_c^{ic*})^\top \Delta \varphi_c^{ic} (W_c^{ic*})^\top \Pi_c^{ic} \nabla \varphi_c^{ic\top} \tilde{W}_a^{ic} \\ &\quad + k_c^{ic} (W_c^{ic*})^\top \tilde{W}_c^{ic} - \frac{1}{2} \tilde{W}_a^{ic\top} \Pi_c^{ic} z_i \end{aligned} \quad (41)$$

Step 3: Differentiating $\mathcal{V}_3$ yields $\dot{\mathcal{V}}_i^3 = -\tilde{W}_a^{ia\top} \Gamma_a^{ia-1} \dot{\tilde{W}}_a^{ia} - \tilde{W}_c^{ia\top} \Gamma_c^{ia-1} \dot{\tilde{W}}_c^{ia}$. By utilizing (25) and (27), one obtains

$$\begin{aligned} \dot{\mathcal{V}}_i^3 &\leq -k_a^{ia} \tilde{W}_a^{ia\top} \tilde{W}_a^{ia} - k_c^{ia} \tilde{W}_c^{ia\top} \tilde{W}_c^{ia} - \frac{1}{2} \tilde{W}_a^{ia\top} \Pi_c^{ia} z_i \\ &\quad - \frac{h}{4m_{ia}} \tilde{W}_a^{ia\top} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \hat{W}_a^{ia} \Delta \varphi_c^{ia\top} \hat{W}_c^{ia} \\ &\quad + k_a^{ia} (W_c^{ia*})^\top \tilde{W}_a^{ia} \\ &\quad + k_c^{ia} (W_c^{ia*})^\top \tilde{W}_c^{ia} + \frac{1}{m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ia} p_{ia} \end{aligned}$$

where $m_{ia} = (1 + \Delta \varphi_c^{ia\top} \Delta \varphi_c^{ia})^2$ and $k_c^{ia} = \|\Delta \varphi_c^{ia}\|^2/m_{ia}$. And then, combining the expressions for $\varepsilon_{hjb}^{ia} = -z_i^\top Q_i z_i + (\tau_{ia}^*)^\top G_i \tau_{ia}^* + \hat{\tau}_{ic}^\top L_i \tau_{ia}^* + (W_c^{ia*})^\top \nabla \varphi_c^{ia} [h_i(z_i) + g_i \hat{\tau}_{ic} + g_i \tau_{ia}^*]$ and $p_{ia} = \int_{t-h}^t r_{ia}(z_i, \hat{\tau}_{ia}, \hat{\tau}_{ic}) ds$, the last term

$\tilde{W}_c^{ia\top} \Delta \varphi_c^{ia} p_{ia}/m_{ia}$ can be given as

$$\begin{aligned} & \frac{1}{m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ic} p_{ia} \\ &= \frac{1}{m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ic} [\bar{h}_{ia} - \int_{t-h}^t \frac{1}{2} (W_c^{ia*})^\top D_c^{ia} \tilde{W}_a^{ic} ds \\ & \quad + \int_{t-h}^t [\hat{\tau}_{ia}^\top G_i \hat{\tau}_{ia} - (\tau_{ia}^*)^\top G_i \tau_{ia}^* - \hat{\tau}_{ic}^\top L_i \tilde{\tau}_{ia}] ds \\ &= \frac{\bar{h}_{ia}}{m_{ia}} \Delta \varphi_c^{ia\top} \tilde{W}_c^{ia} - \frac{h}{2m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ic} (W_c^{ia*})^\top D_c^{ia} \tilde{W}_a^{ic} \\ & \quad - \frac{h}{2m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ia} W_a^{ia*} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ & \quad + \frac{h}{4m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ia} \tilde{W}_a^{ia\top} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \end{aligned}$$

with $\bar{h}_{ia} = \int_{t-h}^t (\varepsilon_{hjb}^{ia} - (W_c^{ia*})^\top \nabla \varphi_c^{ia} [h_i(z_i) + g_i \tau_{ic}^* + g_i \tau_{ia}^*]) ds$, $D_c^{ia} = \nabla \varphi_c^{ia} g_i R_i^{-1} \Psi_i g_i^\top \nabla \varphi_c^{ic\top}$ and

$$\begin{aligned} & \frac{h}{4m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ia} \tilde{W}_a^{ia\top} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ &= -\frac{h}{4m_{ia}} (W_c^{ia*})^\top \Delta \varphi_c^{ia} (W_c^{ia*})^\top \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ & \quad + \frac{h}{4m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ia} (W_c^{ia*})^\top \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ & \quad + \frac{h}{4m_{ia}} (W_c^{ia*})^\top \Delta \varphi_c^{ia} \tilde{W}_a^{ia\top} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ & \quad + \frac{h}{4m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ia} \tilde{W}_a^{ia\top} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \end{aligned}$$

By using the formula obtained from Step 3, $\dot{\mathcal{V}}_i^3$ can be rewritten as

$$\begin{aligned} \dot{\mathcal{V}}_i^3 \leq & -k_a^{ia} \tilde{W}_a^{ia\top} \tilde{W}_a^{ia} - k_c^{ia} \tilde{W}_c^{ia\top} \tilde{W}_c^{ia} + \frac{\bar{h}_{ia}}{m_{ia}} \Delta \varphi_c^{ia\top} \tilde{W}_c^{ia} \\ & + k_a^{ia} (W_c^{ia*})^\top \tilde{W}_a^{ia} + k_c^{ia} (W_c^{ia*})^\top \tilde{W}_c^{ia} - \frac{1}{2} \tilde{W}_a^{ia\top} \Pi_c^{ia} z_i \\ & - \frac{h}{4m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ic} (W_c^{ia*})^\top \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ & - \frac{h}{4m_{ia}} (W_c^{ia*})^\top \Delta \varphi_c^{ia} (W_c^{ia*})^\top \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ & + \frac{h}{4m_{ia}} (W_c^{ia*})^\top \Delta \varphi_c^{ia} \tilde{W}_a^{ia\top} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \tilde{W}_a^{ia} \\ & - \frac{h}{2m_{ia}} \tilde{W}_c^{ia\top} \Delta \varphi_c^{ic} (W_c^{ia*})^\top D_c^{ia} \tilde{W}_a^{ic} \end{aligned} \quad (42)$$

Step 4: Collecting the results in (40), (41), and (42), $\dot{\mathcal{V}}_i$ is derived as

$$\begin{aligned} \dot{\mathcal{V}}_i \leq & -k_{ei} e_i^\top e_i - k_{zi} z_i^\top z_i - k_{ih} \text{tr}(\tilde{W}_{ih}^\top \tilde{W}_{ih}) \\ & - k_c^{ic} \tilde{W}_c^{ic\top} \tilde{W}_c^{ic} - k_a^{ic} \tilde{W}_a^{ic\top} \tilde{W}_a^{ic} - k_c^{ia} \tilde{W}_c^{ia\top} \tilde{W}_c^{ia} \\ & - k_a^{ia} \tilde{W}_a^{ia\top} \tilde{W}_a^{ia} + M_i^1 \tilde{W}_c^{ic} + M_i^2 \tilde{W}_a^{ic} + M_i^3 \tilde{W}_c^{ia} \end{aligned}$$

$$\begin{aligned} & + M_i^4 \tilde{W}_a^{ia} + \tilde{W}_c^{ic\top} M_i^5 \tilde{W}_a^{ic} + \tilde{W}_c^{ia\top} M_i^6 \tilde{W}_a^{ia} \\ & + \tilde{W}_a^{ic\top} M_i^7 \tilde{W}_a^{ic} + \tilde{W}_a^{ia\top} M_i^8 \tilde{W}_a^{ia} \\ & + \tilde{W}_c^{ia\top} M_i^9 \tilde{W}_a^{ic} + k_{ih} b_w^{ih} \|\tilde{W}_{ih}\|_F + b_{zi} \|z_i\| \end{aligned} \quad (43)$$

where $k_{ei} = \lambda_{\min}(K_{ei}) > 0$, $k_{zi} = \lambda_{\min}(K_{zi}) > 0$, and

$$\begin{aligned} M_i^1 &= \frac{\bar{h}_{ic}}{m_{ic}} \Delta \varphi_c^{ic\top} + k_c^{ia} (W_c^{ia*})^\top \\ M_i^2 &= k_a^{ic} (W_c^{ic*})^\top - \frac{h}{4m_{ic}} (W_c^{ic*})^\top \Delta \varphi_c^{ic} (W_c^{ic*})^\top \Pi_c^{ic} \nabla \varphi_c^{ic\top} \\ M_i^3 &= \frac{\bar{h}_{ia}}{m_{ia}} \Delta \varphi_c^{ia\top} + k_c^{ia} (W_c^{ia*})^\top \\ M_i^4 &= k_a^{ia} (W_c^{ia*})^\top - \frac{h}{4m_{ia}} (W_c^{ia*})^\top \Delta \varphi_c^{ia} (W_c^{ia*})^\top \Pi_c^{ia} \nabla \varphi_c^{ia\top} \\ M_i^5 &= \frac{h}{4m_{ic}} \Delta \varphi_c^{ic} (W_c^{ic*})^\top \Pi_c^{ic} \nabla \varphi_c^{ic\top} - \frac{h}{2m_{ic}} D_c^{ic} \\ M_i^6 &= -\frac{h}{4m_{ia}} \Delta \varphi_c^{ic} (W_c^{ia*})^\top \Pi_c^{ia} \nabla \varphi_c^{ia\top} \\ M_i^7 &= \frac{h}{4m_{ic}} (W_c^{ic*})^\top \Delta \varphi_c^{ic} \Pi_c^{ic} \nabla \varphi_c^{ic\top} \\ M_i^8 &= \frac{h}{4m_{ia}} (W_c^{ia*})^\top \Delta \varphi_c^{ia} \Pi_c^{ia} \nabla \varphi_c^{ia\top} \\ M_i^9 &= -\frac{h}{2m_{ia}} \Delta \varphi_c^{ic} (W_c^{ia*})^\top D_c^{ai} \end{aligned}$$

with $\|M_i^j\| \leq m_i^j$, $\forall j = 1, 2, \dots, 9$ for positive constants m_i^j .

By applying Young's inequality, it further yields that

$$\begin{aligned} M_i^1 \tilde{W}_c^{ic} &\leq \frac{k_c^{ic}}{2} \tilde{W}_c^{ic\top} \tilde{W}_c^{ic} + \frac{(m_i^1)^2}{2k_c^{ic}} \\ M_i^2 \tilde{W}_a^{ic} &\leq \frac{k_a^{ic}}{2} \tilde{W}_a^{ic\top} \tilde{W}_a^{ic} + \frac{(m_i^2)^2}{2k_a^{ic}} \\ M_i^3 \tilde{W}_c^{ia} &\leq \frac{k_c^{ia}}{2} \tilde{W}_c^{ia\top} \tilde{W}_c^{ia} + \frac{(m_i^3)^2}{2k_c^{ia}} \\ M_i^4 \tilde{W}_a^{ia} &\leq \frac{k_a^{ia}}{2} \tilde{W}_a^{ia\top} \tilde{W}_a^{ia} + \frac{(m_i^4)^2}{2k_a^{ia}} \\ \tilde{W}_c^{ic\top} M_i^5 \tilde{W}_a^{ic} &\leq \frac{(\kappa_i^5)^2 m_i^5}{2} \tilde{W}_c^{ic\top} \tilde{W}_c^{ic} + \frac{m_i^5}{2(\kappa_i^5)^2} \tilde{W}_a^{ic\top} \tilde{W}_a^{ic} \\ \tilde{W}_c^{ia\top} M_i^6 \tilde{W}_a^{ia} &\leq \frac{(\kappa_i^6)^2 m_i^6}{2} \tilde{W}_c^{ia\top} \tilde{W}_c^{ia} + \frac{m_i^6}{2(\kappa_i^6)^2} \tilde{W}_a^{ia\top} \tilde{W}_a^{ia} \\ \tilde{W}_c^{ia\top} M_i^9 \tilde{W}_a^{ic} &\leq \frac{(\kappa_i^9)^2 m_i^9}{2} \tilde{W}_c^{ia\top} \tilde{W}_c^{ia} + \frac{m_i^9}{2(\kappa_i^9)^2} \tilde{W}_a^{ic\top} \tilde{W}_a^{ic} \\ k_{ih} b_w^{ih} \|\tilde{W}_{ih}\|_F &\leq \frac{k_{ih}}{2} \text{tr}(\tilde{W}_{ih}^\top \tilde{W}_{ih}) + \frac{k_{ih} (b_w^{ih})^2}{2} \\ b_{zi} \|z_i\| &\leq \frac{k_{zi}}{2} z_i^\top z_i + \frac{(b_{zi})^2}{2k_{zi}} \end{aligned} \quad (44)$$

Substituting (44) into (43), it can be deduced as

$$\dot{\mathcal{V}}_i \leq -a_{\mathcal{V}_i} \mathcal{V}_i + b_{\mathcal{V}_i} \quad (45)$$

where $b_{v_i} = (m_i^1)^2/(2k_c^{ic}) + (m_i^2)^2/(2k_a^{ic}) + (m_i^3)^2/(2k_c^{ia}) + (m_i^4)^2/(2k_a^{ia}) + k_{ih}(b_w^{ih})^2/2 + (b_{zi})^2/(2k_{zi})$ and

$$a_{v_i} = \min\{2k_{ei}, k_{zi}, \frac{k_{ih}}{\lambda_{\min}(\Gamma_{ih}^{-1})}, \frac{k_a^{ic} - m_i^5/(\kappa_i^5)^2 - m_i^9/(\kappa_i^9)^2 - m_i^7}{\lambda_{\min}(\Gamma_a^{ic-1})}, \frac{k_c^{ic} - (\kappa_i^5)^2 m_i^5}{\lambda_{\min}(\Gamma_c^{ic-1})}, \frac{k_a^{ia} - m_i^6/(\kappa_i^6)^2 - m_i^8}{\lambda_{\min}(\Gamma_a^{ia-1})}, \frac{k_c^{ia} - (\kappa_i^6)^2 m_i^6 - (\kappa_i^9)^2 m_i^9}{\lambda_{\min}(\Gamma_c^{ia-1})}\}$$

with appropriate finite constants κ_i^5 , κ_i^6 , and κ_i^9 .

Notice that

$$\mathcal{V}_i(t) \leq [\mathcal{V}_i(0) - \frac{b_{v_i}}{a_{v_i}}]e^{-a_{v_i}t} + \frac{b_{v_i}}{a_{v_i}} \leq \mathcal{V}_i(0) - \frac{b_{v_i}}{a_{v_i}}$$

From the definition of $\mathcal{V}_i$ in (39), it can be immediately concluded that the errors e_i , z_i , $\tilde{W}_{ih}^{ic}$, $\tilde{W}_a^{ic}$, $\tilde{W}_c^{ic}$, $\tilde{W}_a^{ia}$, and $\tilde{W}_c^{ia}$ in the closed-loop system remains UUB. According to the control objective (4), define the distributed surrounding error as $\bar{e}_{it} = \eta_i - \eta_t - p_{it}$. The error satisfies that $\|\bar{e}_{it}\| \leq \|e_i\| + \|\tilde{\eta}_{it}\| + 2\rho \sin(\|\tilde{\eta}_{it}\|/2)$. According to Lemma 1 under Assumption 1, $\tilde{\eta}_{it}$ is UUB. Therefore, there exists a positive constant σ_{it} such that the control objective (4) is achieved.

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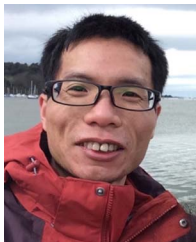
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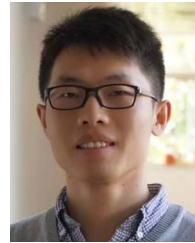
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